

[HOME](#) [TOP](#) [CATALOG](#) [CONTESTS](#) [GYM](#) [PROBLEMSET](#) [GROUPS](#) [RATING](#) [EDU](#) [API](#) [CALENDAR](#) [HELP](#)
[PROBLEMS](#) [SUBMIT CODE](#) [MY SUBMISSIONS](#) [STATUS](#) [STANDINGS](#) [CUSTOM INVOCATION](#)

G. Sorting Again??

time limit per test: 1 second

memory limit per test: 256 megabytes

Forbidden Keywords for the Quiz: `sort`, `open`, `file`, `creat`(, `fstream`, `thread`, `process`, `system`(, `exec`(

Suppose you are given a task to rank the students. You have gotten the marks and ID of the students. Now your task is to rank the students based on their marks using a sorting algorithm. If two or more students get the same mark, then students with the lower ID will get prioritized. See the input and output for a better understanding.

However, **you have to keep in mind that your sorting algorithms perform the minimum number of swapping operations.**

Input

The first line contains an integer T ($1 \leq T \leq 100$) denoting the number of test cases.

For each test case:

The first line contains an integer N ($1 \leq N \leq 1000$).

The second line contains N distinct integers S_i ($1 \leq S_i \leq 1000$), where S_i denotes the student ID.

The third line contains N integers S_m ($1 \leq S_m \leq 1000$), where S_m denotes the obtained mark of the corresponding student.

Note: It is guaranteed that all student IDs are unique.

Output

For each test case, the output must contain a number X which denotes the number of minimum swaps. The rest of the N lines will contain the Student ID and obtained marks sorted based on the instruction above. See the sample output for a better understanding.

Important Note: Since you are asked to minimize the number of swaps, if your number of swaps doesn't match with the judge's answer, your solution will be considered incorrect.

TO CHANGE: Look at the first testcase of the sample input. It can be shown that this can be sorted with only 4 swaps. It can also be shown that it is not possible to sort this in less than 4 swaps.

Example

input	Copy
<pre>2 7 7 4 9 3 2 5 1 40 50 50 20 10 10 10 3 1 2 3 40 59 10</pre>	
output	Copy
<pre>Minimum swaps: 4 ID: 4 Mark: 50 ID: 9 Mark: 50</pre>	

CSE221 Assignment 01 Summer 2026

Contest is running

2 days

Contestant



→ Submit?

Language: Java 21 64bit Choose file: Choose File No file chosen

Success!



Submit

```
ID: 7 Mark: 40
ID: 3 Mark: 20
ID: 1 Mark: 10
ID: 2 Mark: 10
ID: 5 Mark: 10
Minimum swaps: 1
ID: 2 Mark: 59
ID: 1 Mark: 40
ID: 3 Mark: 10
```

[Codeforces](#) (c) Copyright 2010-2026 Mike Mirzayanov
The only programming contests Web 2.0 platform
Server time: Jun/26/2026 10:56:22^{UTC+6} (g1).
Desktop version, switch to [mobile version](#).
[Privacy Policy](#) | [Terms and Conditions](#)

Supported by

